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Yerba Mate Tea (*Ilex paraguariensis*): A Comprehensive Review on Chemistry, Health Implications, and Technological Considerations

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ABSTRACT: Yerba Mate tea, an infusion made from the leaves of the tree *Ilex paraguariensis*, is a widely consumed nonalcoholic beverage in South America which is gaining rapid introduction into the world market, either as tea itself or as ingredient in formulated foods or dietary supplements. The indigenous people have used it for centuries as a social and medicinal beverage. Yerba Mate has been shown to be hypocholesterolemic, hepatoprotective, central nervous system stimulant, diuretic, and to benefit the cardiovascular system. It has also been suggested for obesity management. Yerba Mate protects DNA from oxidation and *in vitro* low-density lipoprotein lipoperoxidation and has a high antioxidant capacity. It has also been reported that Yerba Mate tea is associated to both the prevention and the cause of some types of cancers. Yerba Mate has gained public attention outside of South America, namely the United States and Europe, and research on this tea has been expanding. This review presents the usage, chemistry, biological activities, health effects, and some technological considerations for processing of Yerba Mate tea. Furthermore, it assesses in a concise and comprehensive way the potential of *Ilex paraguariensis* as a source of biological compounds for the nutraceutical industry.

Keywords: *Ilex paraguariensis*, Mate tea, nutraceutical industry, tea, Yerba Mate

Introduction

Yerba Mate tea (Mate), an herbal tea beverage widely consumed in southern Latin American countries (southern Brazil, Argentina, Paraguay, and Uruguay) is gaining rapid penetration into world markets, including the United States. It is made from an infusion of the dried leaves of *Ilex paraguariensis*, a plant of the Aquifoliaceae family (Small and Catling 2001; Grigioni and others 2004). In Latin America, Mate is often druank out of a dried gourd using a metal straw called "bombilla." The dry leaves (about 50 g) are packed into the gourd and hot water is poured over them; this is then repeated multiple times, with as much as half to 1 L of water. In the United States, however, Mate is commercially packed in individual tea bags (1 to 2 g) or as Mate tea concentrate for use as ingredient in the food or dietary supplement industries. Considering the importance of the growing consumption of Mate tea and Mate tea containing products, the objective of this review is to compile and comprehensively analyze updated scientific information on Yerba Mate, including its composition, physiological effects, and potential health implications. In addition, this review hopes to further stimulate uses of Yerba Mate as nutraceutical ingredient. This compiled knowledge may provide a central resource for future research on Yerba Mate.

Mate tea has recently been highly publicized for its health benefits but there have been also concerns about its safety. The scientific literature, on one hand, reports that Mate tea is hypocholesterolemic, hepatoprotective (Filip and Ferraro 2003), central nervous system stimulant, diuretic (Gonzalez and others 1993), and antioxidant (Filip and others 2000; VanderJagt and others 2002). It also has benefits to the cardiovascular system (Schinella and others 2005), and is a protector of DNA oxidation and *in vitro* low-density lipoprotein (LDL) lipoperoxidation (Bracesco and others 2003). Some studies have also suggested its potential in the management of obesity (Andersen and Fogh 2001; Pittler and Ernst 2004; Opala and others 2006). Numerous active phytochemicals have been identified in Mate tea that may be responsible for its health benefits. Among them, the 2 highest compounds are the polyphenols (chlorogenic acid) and xanthines (caffeine and theobromine), followed by purine alkaloids (caffeic acid, 3, 4-dicaffeoylquinic acid, 3, 5-dicaffeoylquinic acid), flavonoids (quercetin, kaempferol, and rutin), amino acids, minerals (P, Fe, and Ca), and vitamins (C, B1, and B2) (Pomilio and others 2002; Zaporozhets and others 2004). Not only has Mate tea been shown to contain high concentrations of bioactive compounds, it has also been shown to be cytotoxic to human cancer hepatoma cells (HepG2), and can act as a catalytic inhibitor of topoisomerase II (Ramirez-Mares and others 2004).

On the other hand, some epidemiological studies have reported an association between the consumption of Mate tea and an increased risk of various types of cancer, including oral, oropharyngeal, esophageal, laryngeal, and bladder (Goldenberg and others 2003; Sewram and others 2003; Bates and others 2007).

Ethnobotany and Botanical Description

Ilex paraguariensis, from the family of holy plants, Aquifoliaceae, is a native South American tree used for the production of Yerba Mate tea. It is found primarily in the southern regions of South America, namely, Brazil (Mato Grosso do Sul, Minas Gerais, Parana, Rio Grande do Sul, Rio de Janeiro, Santa Catarina, Sao Paulo), Argentina (Corrientes, Misiones), Paraguay (Alto Parana, Amambay, Caaguazu, Canendiyu, Central, Guaira, Itapua, Misiones, San Pedro), and Uruguay (USDA, ARS, National Genetic Resources Program 2007). Figure 1A shows the main regions where Mate is

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grown. Of these regions, Argentina is the largest producer, cultivating around 152000 hectares of Mate per year in the northeastern part of the country (Misiones and Corrientes). This is equal to approximately 280000 tons per year, representing a large portion of the countries gross domestic product. Brazil and Paraguay are the 2nd and 3rd largest producers, respectively. Worldwide, 290000 Ha of area harvested with a production of 874678 tons of Mate were reported in 2002 (FAOSTAT 2007). The overall value of Mate production around the world is estimated in U.S. $1 billion in 2004.

Ilex paraguariensis is a subtropical dioecious evergreen tree that can reach 18 m in height. Figure 1B shows a picture of the Mate plant. The Mate tree is a flower and fruit producing plant, flowering from October to November and producing fruit from March to June. The Mate plant requires a strict regimen of annual rainfall both in amount, no less than 1200 mm, and distribution throughout the year. It is, however, much less susceptible to temperature, being able to withstand temperatures of –6 °C, with an average annual temperature of 21 to 22 °C. It is also able to withstand the frequent snowfalls that are attributed to the mountainous region in which it inhabits.

The cultivation and harvesting of Mate is not a uniform procedure and is conducted by various methods depending on the region. The 3 primary ways for cultivation and harvest are extractive exploitation of the natural forest, mixed system, and cultivated Mate plantations. Extractive exploitation of the natural forest utilizes wild harvesting of Mate from the forest and is the most inconsistent of the 3 methods based on quality and quantity. The 2nd method, mixed system, combines forest growth with better cultivation practices, including replanting of plants as they are lost and improved pruning methods. This practice yields a better production rate for the growing of natural forest products. Both natural forest harvest and mixed system cultivation are primarily found in Brazil. Cultivated Mate plantations, believed to be the most efficient method of production, began in Argentina since 1915. This method increases both yield and harvest efficiency by allowing for growth of more plants in a given area and the use of mechanical harvesting, which offsets the higher growing cost (Giberti 1994).

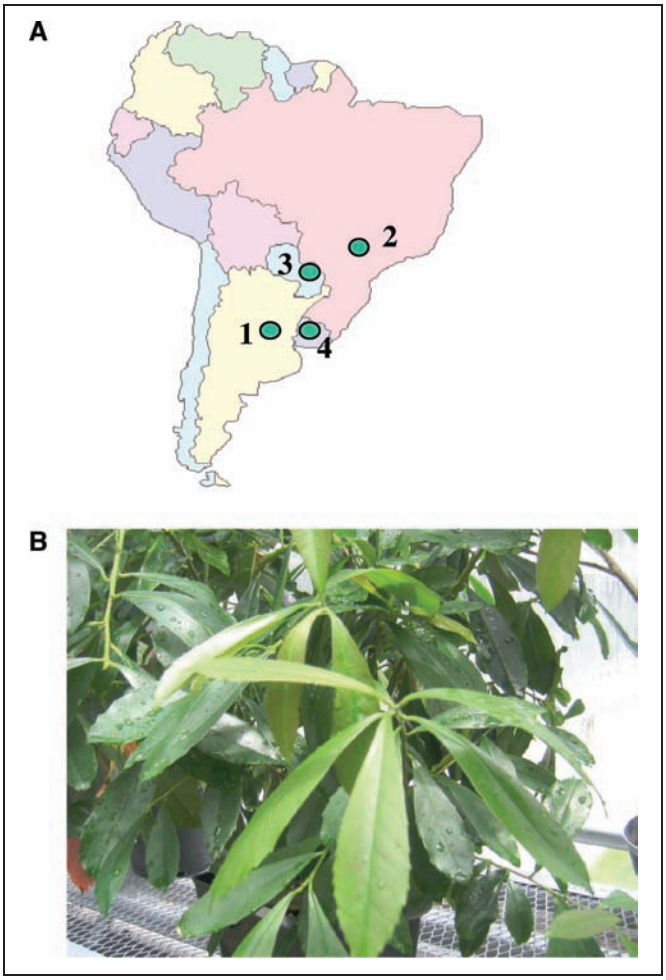

Figure 1 – (A) Map of South America showing growing regions for Yerba Mate *(Ilex paraguariensis)* 1 Argentina; 2 Brazil, 3 Paraguay, 4 Uruguay. (B) Yerba Mate plant.

Mate Tea Processing

Yerba Mate is not consumed as a raw product but instead it is processed before it reaches the consumer. Fresh Mate leaves undergo several stages of processing before it is ready to be packaged. This involves blanching, drying, and generally aging of the tea. The conditions for processing are widely varied depending on the producer and the final objective for the desired style and flavor of Mate tea. Processors can vary the time and temperature of blanching and drying. Not all producers will age the tea, while others will vary the aging time (Bastos and others 2006a). However, the overall process is generally the same. Figure 2A shows a typical process flow chart for Mate tea.

Mate goes through very little fermentation and the blanching process that deactivates enzymes, that is, polyphenol oxidase. The difference in the blanching process, however, is that green tea leaves are steamed or pan-fried and Mate tea leaves are flash heated over open flame. This blanching process is in contrast to that for the production of black tea; the leaves for black tea are allowed to wither and ferment and are not blanched before drying. Figure 2B shows the process for producing green and black teas. In black tea, the enzyme polyphenol oxidase is allowed to oxidize polyphenols to form dimerized compounds, that is, catechins to theaflavins (Hara 2001).

The major difference between green tea and Mate tea production is the drying method. Green tea is dried primarily through a fast, high temperature air drying, which retains more of the fresh leave characteristics, as well as developing characteristic flavor and aroma compounds. Mate tea is dried very slowly and often using wood smoke. This imparts very different flavor characteristics and contributes to changes in the chemical makeup and physical appearance. Another important difference between Mate and green tea is the presence of stems in the final product. Green tea production removes all large stems before grinding (Graham 1992); however, Mate will generally have a high content of stem pieces present, depending on the producer.

Phytochemistry

Polyphenols

Polyphenols are a class of compounds containing a benzene ring bound with one or more hydroxyl groups. These compounds have been analyzed with a number of methods, including a tyrosinase biosensor, Folin Ciocalteu assay, and high-performance liquid chromatography (HPLC) (Carini and others 1998; Chandra and De Mejia Gonzalez 2004; Dall'Orto 2005). With these analyses it has been shown that the variety of Mate, degree of milling, and blending with other teas determine the concentration of polyphenols extracted in an infusion. On average, the amount of polyphenols extracted from Mate tea is 92 mg equivalents of chlorogenic acid per gram of dry leaves, with blended teas having significantly less (Dall'Orto 2005). The polyphenol concentration of Mate has also shown a

strong correlation to its overall antioxidant capacity (Chandra and De Mejia Gonzalez 2004). Mate showed a slightly higher polyphenol concentration, 7.73 ± 0.15 mg chlorogenic acid/mL water extract, than green tea, 7.15 ± 0.14 mg chlorogenic acid/mL water extract. This correlates to a higher antioxidant capacity for Mate, 90.45 ± 0.22% inhibition of free radical, than green tea, 88.36 ± 0.76% inhibition of free radical, when the 1,1-diphenyl-2-picrylhydrayl (DPPH) method was used (Bastos and others 2007). Furthermore, the amount of polyphenols extracted from Mate is affected by the extraction method used, that is, water or organic solvent, with 50% acetone extraction yielding the highest amount of polyphenols (Turkmen and others 2006).

Polyphenolic compounds found in Mate tea differ significantly from green tea because Mate tea contains high concentration of chlorogenic acid and no catechins (Chandra and De Mejia Gonzalez 2004). Table 1 shows the diversity of polyphenolic compounds in green tea, black tea, and Mate tea.

Xanthines

Xanthines are a class of purine alkaloids found in many different plants, including tea, coffee, and chocolate. The xanthines found in Mate include theophylline (1,3-dimethylxanthine), theobromine (3,7-dimethylxanthine), and caffeine (1,3,7-trimethylxanthine) (Athayde and others 2000). The structural formulas of these compounds are presented in Figure 3. Of these three, caffeine is found in the highest concentration, 1% to 2% of dry weight, followed by theobromine, 0.3% to 0.9% of dry weight (Ito and others 1997). These 2 compounds are found primarily in the leaves of the plant and in smaller concentrations in the woody stems that are often present in the product as well as in the epicuticular waxes of the leaves (0.5% wax content of dry leaf weight), with 5.9 to 17.0 ng of caffeine per milligram of wax and 0.9 to 3.5 ng theobromine per milligram of wax (Athayde and others 2000), though the major quantities of these methylxanthines exist inside the leaves.

The concentration of caffeine in relation to consumer consumption has been found to be approximately 78 mg of caffeine in 1 cup of Mate tea (approximately 150 mL). Compared to coffee, this is a very similar amount of caffeine consumption, approximately 85 mg per cup. However, the customary rate of Mate consumption prepared in the traditional method can present intakes of around 500 mL, resulting in 260 mg or more of total caffeine (Mazzafera 1997).

In contrast to theobromine and caffeine, theophylline has been found in only small quantities in the leaves. This may be due to the fact that theophylline appears to be an intermediate in the catabolism of caffeine in the plant. It is believed that the main route of theophylline metabolism involves conversion to 3-methylxanthine, which is further demethylated to xanthine prior to entering the purine catabolism pathway and being degraded via a xanthine → uric acid → allantoin → allantoic acid → → $CO_2 + NH_3$ route. It has been shown that when theophylline is radioactively labeled, the label will show up in caffeine and theobromine through the resynthesis of caffeine via a theophylline → 3-methylxanthine → theobromine → caffeine pathway (Ito and others 1997). The fact that theophylline has been difficult to find in varying tests on Mate may be due to theophylline metabolism into caffeine and theobromine.

Yerba Mate is often sold as dried ground leaves; however, it has been suggested that the drying process can significantly affect the concentration of caffeine as well as color and chlorophyll content of the leaves. Schmalko and others (2001) examined the caffeine, color, and chlorophyll content of Mate leaves after 3 stages of drying. The 1st stage was blanching, sapeco, with a temperature of 500 to 550 °C for 2 to 4 min; the 2nd and 3rd stages were the drying stages, barbaqua, with a temperature of about 110 °C. These drying stages showed a dramatic decrease in caffeine (30%) and

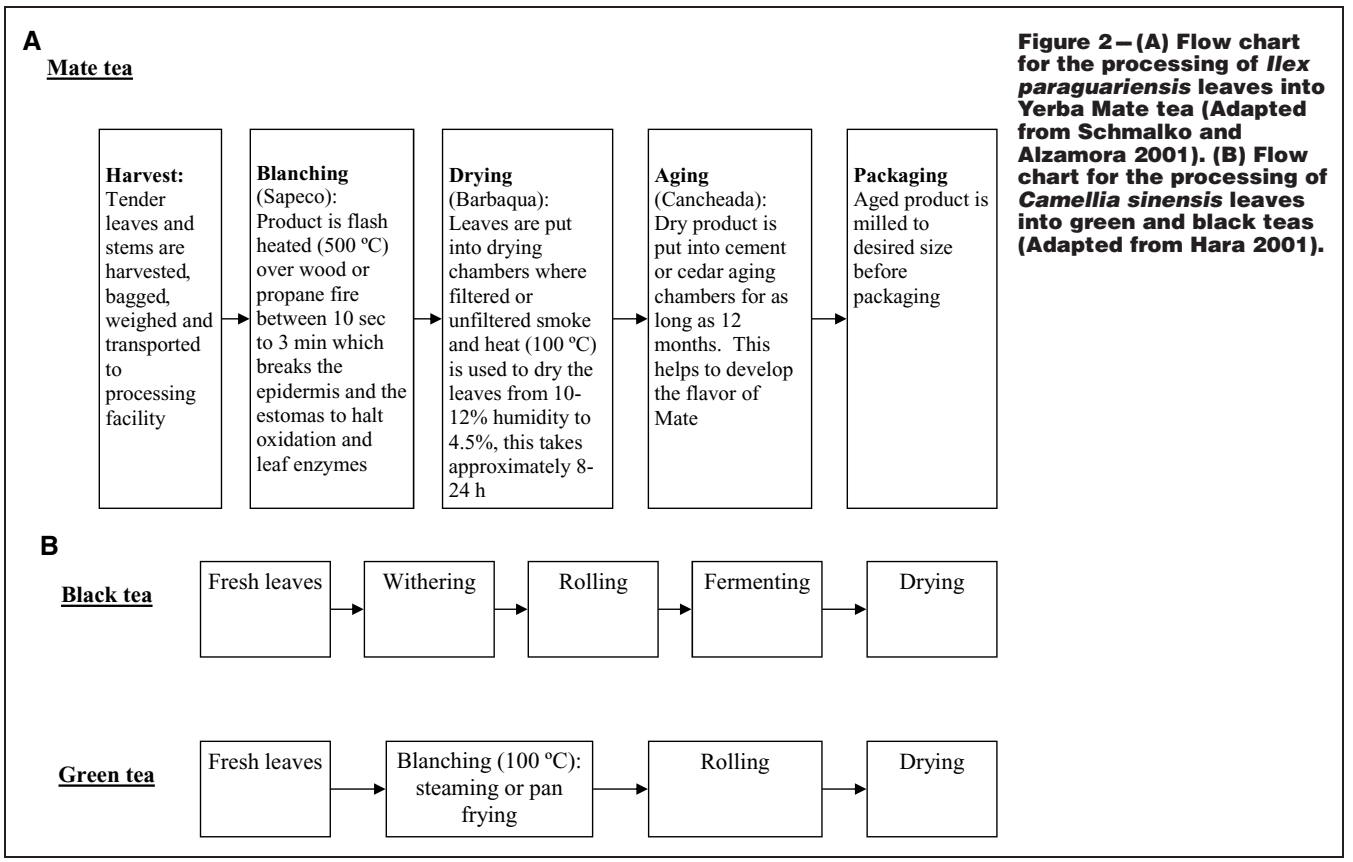

Figure 2—(A) Flow chart for the processing of *Ilex paraguariensis* leaves into Yerba Mate tea (Adapted from Schmalko and Alzamora 2001). (B) Flow chart for the processing of *Camellia sinensis* leaves into green and black teas (Adapted from Hara 2001).

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chlorophyll (70% to 80%) concentrations, and a decrease of green color. However, even though the caffeine concentration in the dried product was lower than in fresh leaves, evidence by Bastos and others (2006a showed that when the leaves were dried and used to prepare Mate infusions, significantly more caffeine and caffeoylquinic acids were extracted than when using fresh leaves. This increased extraction of compounds is likely from the disruption of the cells during the drying process. It may also be explained by a decrease in moisture concentration to leaves and an increase in soluble solids during drying, thus leading to a greater amount of compounds dissolved into the infusion. Evidence has also been presented that the time of harvest plays a role in the concentration of methylxanthines found in Mate, ranging between 1 and 10 mg total methylxanthines/g depending on time of harvest (Schubert and others 2006).

Caffeoyl derivatives

The caffeoyl derivatives found in Mate include caffeic acid, chlorogenic acid, 3, 4-dicaffeoylquinic acid, 3, 5-dicaffeoylquinic acid, and 4, 5-dicaffeoylquinic acid (Filip and others 2000). These caffeoyl derivatives are the primary constituents that account for the antioxidant capacity of Mate tea. Figure 4 shows the chemical structure for chlorogenic acid, 4,5-dicaffeoylquinic acid, 3,5-dicaffeoylquinic acid, and 3,4-dicaffeoylquinic acid. They have been analyzed primarily by 2 different methods, spectrophometrically (330 nm) and by HPLC, and are often correlated with chlorogenic acid as a standard with a concentration of 6.90 ± 0.09 mg chlorogenic acid/g dry leaves (Filip and others 2000). This is representative of 0.48 mg chlorogenic acid/mL and roughly 72 mg in 1 cup (150 mL) of Mate brew, when prepared with 1.5 g per 50 mL water (Mazzafera 1997). These compounds can also be identified individually by HPLC and in combination with liquid chromatography/mass spectrometry (LC/MS), with absorptions at 242, 228, and 330 nm (Carini and others 1998; Chandra and De Mejia Gonzalez 2004). Figure 5 shows a chromatographic profile generated by our group for the identification of caffeoyl derivatives in Mate (*I. paraguariensis*) (Heck and Gonzalez de Mejia 2007). It is apparent that the large constituents are chlorogenic acid and its derivatives and the dicaffeoylquinic acids: 3,4-dicaffeoylquinic acid, 3,5-dicaffeoylquinic acid, and 4,5-dicaffeoylquinic acid; though the specific identity of each dicaffeoylquinic acid peak has not been described (Carini and others 1998). This profile agrees with the compounds presented in Table 2 regarding the concentrations of the caffeoyl derivatives found in Mate (*I. paraguariensis*) compared with *I. dumosa, I. brevicuspis,* and *I. argentina*. This table shows that *I. paraguariensis* contains the highest concentrations of the caffeoyl derivatives while the other species have much lower concentrations and vary in their dicaffeoylquinic acid concentrations (Filip and others 2001). It is because of the high concentrations of these compounds that Mate possesses a very high overall antioxidant capacity (Filip and others 2000).

Table 1—Polyphenols in green tea, black tea, and Mate tea.[a]

	Green tea	Black tea	Mate tea
Caffeic acid	•		•
Caffeine	•	•	•
Caffeoyl derivatives			•
Caffeoylshikimic acid			•
Catechin	•	•	
Catechin gallate		•	
Chlorogenic acid			•
Coumaric acid	•		
Epicatechin gallate	•	•	
Epigallocatechin	•		
Epigallocatechin gallate	•		
Feruloylquinic acid			•
Gallic acid	•	•	
Gallocatechin gallate	•	•	
Kaempferol	•	•	•
Myricetin	•	•	
Procyanidin		•	
Quercetin	•	•	•
Quinic acid	•		•
Rutin	•	•	•
Theaflavin		•	
Theobromine		•	•

[a]Adapted from Carini and others (1998); Chandra and de Mejia Gonzalez (2004); Atoui and others (2005); Bastos and others (2007); Bravo and others (2007).

Saponins

Saponins are bitter and highly water-soluble compounds found in many types of plants and they are believed to be one of the factors for the distinct flavor of Mate tea. Not only do they play a role in flavor but are also attributed to anti-inflammatory and hypocholesterolemic properties (Gnoatto and others 2005). Several of these compounds, namely, triterpenoid saponins with ursolic and oleanolic moieties, have been isolated from the leaves of Mate. The primary saponins identified contained the ursolic acid moiety and were named: Matesaponins 1, 2, 3, 4, and 5 (Gosmann and others 1995; Kraemer and others 1996). Table 3 shows the primary saponins identified in Mate (*I. paraguariensis*) as well as those for other species of *Ilex*; included are common R group substitutions. Figure 6 shows a structure of a generic saponin aglycon onto which various R groups are attached. The hypocholesterolemic properties may be attributed to the Mate saponin inhibition of passive diffusion of colic acid and formation of micelles that cannot be absorbed and are thus excreted (Ferreira 1997).

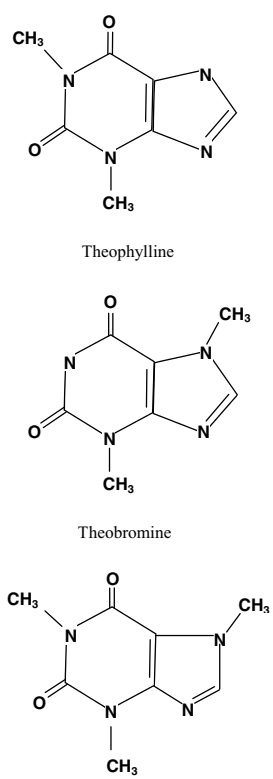

Figure 3—Structure of xanthines: theophylline (1,3-dimethylxanthine), theobromine (3,7-dimethylxanthine), and caffeine (1,3,7-trimethylxanthine).

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Table 2—Concentration of caffeoyl derivatives in various *Ilex* species (% of dried weight).[a]

Species	Chlorogenic Acid	Caffeic Acid	3,4-DCQ	3,5-DCQ	4,5-DCQ
I. paraguariensis	2.800 ± 0.300	0.023 ± 0.004	0.855 ± 0.064	3.040 ± 0.180	2.890 ± 0.060
I. brevicuspis	0.915 ± 0.064	0.005 ± 0.001	0.130 ± 0.010	0.360 ± 0.060	0.490 ± 0.040
I. argentina	0.090 ± 0.015	0.003 ± 0.001	0.047 ± 0.010	0.545 ± 0.049	0.043 ± 0.003
I. dumosa	0.042 ± 0.009	0.012 ± 0.008	0.017 ± 0.001	0.147 ± 0.060	0.070 ± 0.014

[a]Adapted from Filip and others (2001).
3,4-DCQ = 3,4-dicaffeoylquinic acid; 3,5-DCQ = 3,5-dicaffeoylquinic acid; 4,5-DCQ = 4, 5-dicaffeoylquinic acid.

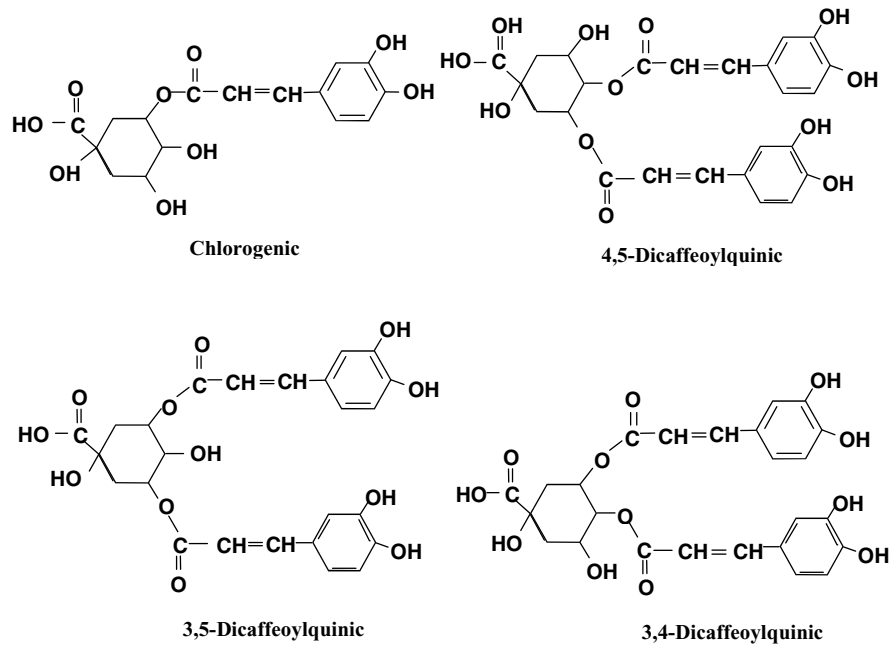

Figure 4—Structure of caffeoyl deriviatives: chlorogenic acid, 4,5-dicaffeoylquinic acid, 3,5-dicaffeoylquinic acid, and 3,4-dicaffeoylquinic acid.

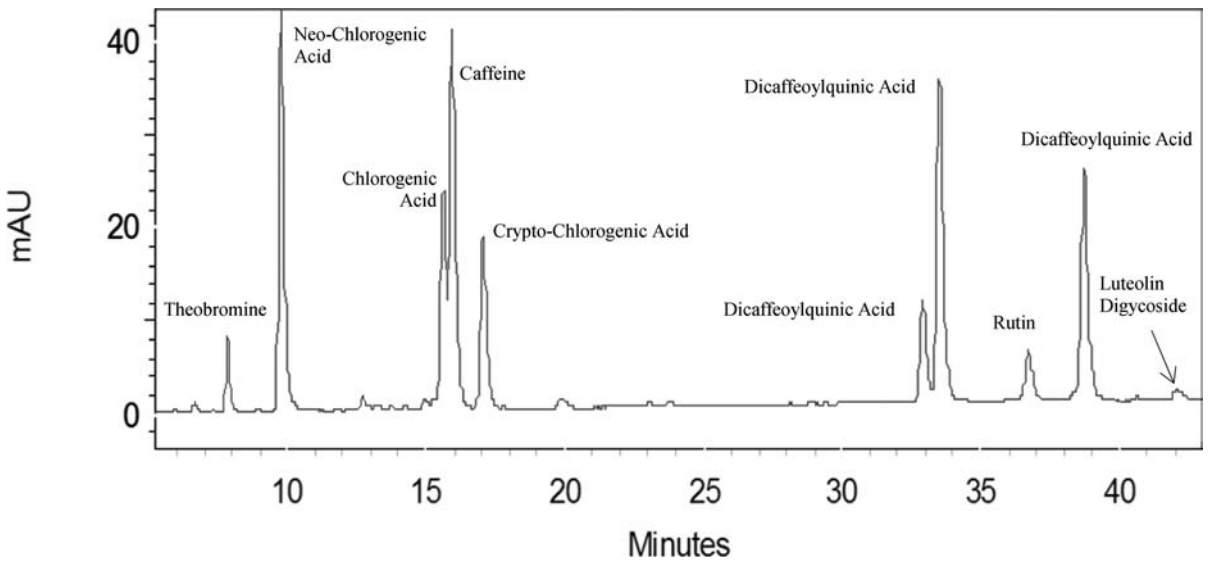

Figure 5—Chromatographic (HPLC) profile of Mate tea identifying caffeoyl derivatives and other compounds (Heck and de Mejia 2007). Analysis was conducted using a 1050 Hewlett Packard (Palo Alto, Calif., U.S.A) gradient liquid chromatograph, equipped with a 1050 HP auto sampler, a 1050 HP gradient pump, a 1050 HP photodiode array detector (PDA), and helium sparge. A C_{18} RP guard column and a C_{18} RP Phenomenex Prodigy ODS column (250 mm × 4.6 mm × 5 μm) were used. Column temperature was kept at ambient temperature, elution time was 0.9 mL/min, and was performed with a solvent gradient. Solvent gradient consisted of solvent A (water/methanol/formic acid, 79.7/20/0.3) and B (methanol/formic acid, 99.7/0.3) mixed, starting with 0% B, linearly increasing to 25% B in 50 min, increase to 80% B in 5 min and held at 80% B for 3 min, then a linear decrease to 0% B in 5 min and held at 0% B for 5 min. Injection volume was 50 μL and output at 280 nm.

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Gnoatto and others (2005) recently developed a method utilizing HPLC with ultraviolet (UV) detection for analysis of saponins in Mate. Total recovery of Matesaponin 1 was 94.5% and total concentration of saponins in the aqueous extract was 352 μg/mL from 15 g of dried leaves in 100 mL water. While the main saponins in Mate are formed with ursolic acid aglycons, 2 minor saponins have also been identified that contain an oleanolic acid instead of the ursolic acid (Martinet and others 2001). Pavei and others (2007) have also developed and validated an HPLC method to characterize saponins from *I. paraguariensis* Mate fruits.

Many of the saponins found in *Ilex* species have been shown to possess antiparasitic properties, including Matesaponins 1, 3, and 4. It has also been confirmed that triterpenoids found in *Ilex* species are antitrypanosomal. Ursolic acid and 4, 3-*O*-[α-D-glucopyranosyl-(1-2)- α-D-galactopyranosyl] oleanolic acid had an IC_{50} of 4 μM against *Trypanosoma brucei*. These findings may lead to the examination of the use of these compounds for new antitrypanosomal derivatives (Taketa and others 2004).

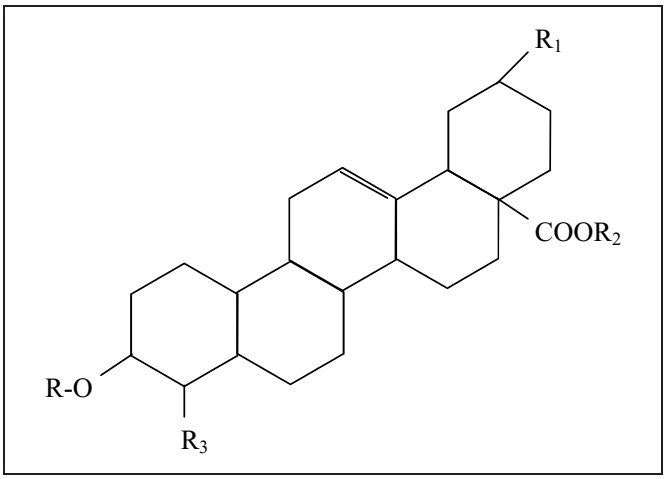

Figure 6—Generic saponin structure with locations of common R group substitutions.

Minerals

Mate also contains high concentrations of inorganic compounds. The minerals aluminum, chromium, copper, iron, manganese, nickel, potassium, and zinc are of particular interest due to their importance in human metabolism and development. Using capillary ion electrophoresis with indirect UV detection (Carducci and others 2000) and atomic absorption spectrophotometry (Tenorio Sanz and Torija Isasa 1991; Vera Garcia and others 1997), these minerals have been identified in varying concentrations and can vary based on soil and seasonal factors. Using particle-induced X-ray emission (PIXE), Giulian and others (2007) assayed Mate tea brands before and after infusion and found a wide range of minerals and that some depend on temperature and volume used in the infusion, namely, chlorine and potassium. Wrobel and others (2000) found the aluminum concentration as 369 ± 22 μg/g and a manganese concentration of 2223 ± 110 μg/g; Mate could prove to be a good dietary source of manganese, depending on bioavailability. It should also be noted that an inverse correlation (correlation coefficients >0.82) was found between the amount of these minerals leached into a Mate infusion and the tannin concentration; in the lower tannin concentrations the best leaching was observed, with the exception of nickel.

In addition to beneficial elements, toxic contaminants could be present in Mate as well. Marchisio and others (2005) developed a lead analysis method using ultrasonic nebulization associated to inductively coupled plasma optical emission spectrometry (USN-ICP-OES) and polyurethane foam. Their method demonstrated a process for the detection of lead that proved to be fast, accurate, and reliable and can measure small concentrations of lead. The concentrations of lead in Mate infusions were in the range between 7.6 and

Table 3—Saponins of *Ilex* species and their structural differences including R group substitutions.

Ilex Species	Saponin	Moiety	R	R1	R2	R3
paraguariensis[a]	Matesaponin 1	Ursolic acid	glc(1→3)ara	H	glc	H
	Matesaponin 2	Ursolic acid	glc(1→3)rha(1→2)ara	H	glc	H
	Matesaponin 3	Ursolic acid	glc(1→3)ara	H	glc(1→6)glc	H
	Matesaponin 4	Ursolic acid	glc(1→3)rha(1→2)ara	H	glc(1→6)glc	H
	Matesaponin 5	Ursolic acid	glc(1→3)rha(1→2)ara	H	glc(1→4)glc(1→6)glc	H
affinis[b]	Affinoside I	Pomolic acid	glc(1→3)ara	H	glc	H
crenata[c]	Ilexoside II	Pomolic acid	glc(1→3)ara	H	glc	H
integra[d]	Ilexoside XXV	Hydroxyursolic acid	glc	H	glc	CH_2OH
	Ilexoside XXVI	Hydroxyursolic acid	glc(1→6)glc	H	glc	CH_2OH
	Ilexoside XXVII	Rotundic acid	ara	H	glc	CH_2OH
buxifolia[b]	Buxifolioside I	Dihydroxyursendioic acid	H	H	glc	CH_3
	Buxifolioside II	Dihydroxyursendioic acid	OH	H	glc	COOH
dumosa[e]	Chikusetsusaponin Iva	Oleanolic acid	gluA	H	glc	H
	Dumosaponin 5	Oleanolic acid	glc(1→2)gal	OH	glc	H
	Dumosaponin 6	Oleanolic acid	ara(1→2)ara	H	glc	H
	Dumosaponin 7	Oleanolic acid	gal	H	glc	H
latifolia[f]	Latifolioside F	Ilexgenin	rha(1→2)glc(1→3)ara	H	rha(1→2)glc	H
	Latifolioside G	Polmolic acid	rha(1→2)glc(1→3)ara	H	rha(1→2)glc	H
	Latifolioside H	Siaresinolic acid	rha(1→2)glc(1→3)ara	H	rha(1→2)glc	H
argentina[g]	N/A	Rotundioic acid	H	H	glc	COOH
rotunda[h]	Ilexosides XXXIII	Oxosiaresinolic acid	GlcA	H	H	CHO
	Ilexosides XXXIV	Pedunculoside	SO_3Na	H	glc	H
	Ilexosides XXXV	Rotungenic acid	SO_3Na	H	glc	CH_2OH
	Ilexosides XXXVI	Rotungenic acid	glc	H	glc	CH_2OH
	Ilexosides XXXVII	Rotundic acid	glc	H	glc	H
brevicuspis[i]	Brevicuspisaponin I	Hydroxyursolic acid	ara	H	H	CH_3
	Brevicuspisaponin II	Hydroxyursolic acid	ara	H	H	CH_2OH

gluA = glucuronic acid; glu = glucose; gal = galcatose; ara = arabinose; rha = rhamnose; SO_3Na = sulfate.
Gnoatto and others 2005[a]; Taketa and others 2004[b]; Taketa 2004[c]; Yano and others 1993[d]; Pires and others 1997[e]; Ouyang and others 1998[f]; Pires and others 2002[g]; Amimoto and others 1993[h]; Taketa and others 2000[i].

8.9 μg/L. The average concentration of lead in commercial Mate tea samples analyzed was 8.1 μg/L. The allowable limit for lead in drinking water by the U.S. Environmental Protection Agency (EPA) is 15 μg/L; therefore, the levels found in Mate are well below the level for concern (EPA 2003).

Mate adulterants

Adulterants of other *Ilex* species may be incorporated into the final product, either intentionally or unintentionally. Six common *Ilex* species found as adulterants in Mate tea were tested for theobromine, theophylline, and caffeine. The species analyzed were *I. dumosa, I. pseudobuxus, I. brevicuspis, I. theezans, I. microdonta,* and *I. argentina;* overall results showed that these other species contained little to none of the aforementioned compounds. Only traces of caffeine were detected in *I. theezans, I. dumosa, I. microdonta,* and *I. pseudobuxus*. Furthermore, only traces of theobromine were detected in *I. argentina* and *I. microdonta*. Theophylline was only quantifiably detected in *I. pseudobuxus* at 6 ppm (Filip and others 1998). Utilizing HPLC and NMR to analyze *Ilex* varieties caffeine and theobromine were only found in *I. paraguariensis* compared to other *Ilex* adulterating species (Reginatto and others 1999; Choi and others 2005).

These adulterants can be problematic for the quality of Mate teas due to their differing concentration of saponins. Mate tea prepared with *I. paraguariensis*, showed to be the least bitter of all extracts prepared with adulterating species. Thus, it is possible that the addition of adulterating species can have a significant effect on the bitterness of Mate beverages. Not only do the adulterating plants contain greater concentrations of bitter compounds but the fruits of the *I. paraguariensis* plant itself also contain highly bitter saponins. If these fruits were incorporated into the Mate products it may lead to an increase in bitterness and a decrease in overall quality (Taketa 2004).

A number of these species have also been analyzed for their saponin concentration. Analysis showed that a majority of the species including *I. buxifolia*, *I. crenata*, *I. affinis, I. rotunda, I. brevicuspis, I. argentina,* and *I. integra* all have saponin aglycons not found in *I. paraguariensis* and *I. dumosa*; instead of ursolic acid or oleanolic acid aglycons, they possess hydroxyursolic acid or derivatives. Of the various *Ilex* species, *I. dumosa* is the most prevalent adulterant and the more similar to *I. paraguariensis* saponin structure. All adulterating species, including *I. dumosa*, contained a large variation in saponins, none of which was found in *I. paraguariensis*. Due to the specificity of saponins, it may be possible to identify adulterants in Mate based on saponin concentration, and with new methods for rapid and precise identification of adulterants this may now be a plausible method for the quality control of Yerba Mate products (Pires and others 1997).

Biological Activities and Health Effects

Table 4 shows an incomplete list of compounds that have been identified in Yerba Mate and some of the most important reported biological activities follow.

Antioxidant capacity

It has been found that the consumption of Mate tea significantly contributes to the overall antioxidant intake and provides high amounts of caffeoylquinic acid derivatives, with biological effects potentially beneficial for human health (Bravo and others 2007). Of all the *Ilex* species, *I. paraguariensis* has been shown to contain the highest antioxidant activity and has been positively correlated with the concentration of caffeoyl derivatives (Filip and others 2000; Schinella and others 2000; Bracesco and others 2003; Bixby and others 2005). The study of Mate's ability to quench reactive oxygen species (ROS) has been correlated to peroxidase-like activity. This peroxidase-like activity is strongly related to the polyphenol concentration of Mate; the higher the polyphenol concentration, the greater the peroxidase-like activity. This means, from the biological standpoint, that polyphenols act similarly as the bodies 293 natural antioxidant enzymes and may prove to be potent supporters of these systems.

The compound that may be primarily responsible for this activity is chlorogenic acid (Anesini and others 2006).

Mate extract has shown to be a very potent inhibitor of oxidative stress caused by ROS, considerably so for the liver and heart. The heart is susceptible to oxidative stress during postischemic reperfusion, return of blood flow to organ and tissue after heart attack, caused by the generation of ROS. Administering Mate extract decreased the lipid oxidation in the heart by protecting myocardial tissue (Schinella and others 2005).

Recent studies have shown that nitrosative stress, a reaction of superoxides with nitrous oxide (NO) forming peroxynitrite (ONOO), causes protein nitration or nitrosylation, lipid peroxidation, DNA damage, and cell death. Mate tea was able to prevent 95% of protein nitration when tested on bovine serum albumin; in this respect, Mate was higher than both green tea and red wine. Mate was also tested against peroxynitrite-induced cytotoxicity, associated with stroke and myocardial ischemia, restriction in blood supply, and Mate tea showed the highest inhibition against cytotoxicity, compared with green tea and red wine (Bixby and others 2005). Mate has also been able to reduce ATP, ADP, and AMP (nucleotide) hydrolysis, which can help balance the circulatory system (Gorgen and others 2005).

Table 4—Compounds identified in Yerba Mate leaves and some of their biological activities.

Compound	Biological activities
Caffeine	Anticarcinogenic, antiobesity, antioxidant, antitumor, diuretic, energizer 20 to 200 mg, stimulant, topoisomerase-I-inhibitor 0.1 M, topoisomerase-II-inhibitor 99 mM, vasodilator
Chlorogenic-acid	Antioxidant $IC_{50} = 54.2$ μM, analgesic, antiatherosclerotic, antibacterial, antidiabetic, antitumor, choleretic
Chlorophyll	Antibacterial, anticancer
Choline	Antidiabetic, cholinergic, lipotropic
Nicotinic acid	Choleretic, hypocholesterolemic 1 to 6 g/day
Pantothenic acid	Antiallergic 100 to 500 mg/day, antiarthritic 500 to 2000 mg/day, antifatigue
Rutin	Antioxidant $IC_{28} = 30$ ppm $IC_{50} = 120$ μM, antitumor, antitumor-promoter, antiulcer, cAMP-phosphodiesterase-inhibitor, topoisomerase-II-inhibitor $IC_{50} = 1$ μg/mL, vasodilator
Tannin	Antioxidant 1/3 quercetin $IC_{50} = 1.44$ μg/mL, antitumor, antitumor-promoter, lipoxygenase-inhibitor, MAO-inhibitor[e]
Theobromine	cAMP-inhibitor $IC_{50} = 0.06$ mg/mL, cAMP-phosphodiesterase-inhibitor, diuretic 300 to 600 mg/day, stimulant, myorelaxant
Theophylline	cAMP-inhibitor $IC_{50} = 0.06$ mg/mL, cAMP-phosphodiesterase-inhibitor, diuretic, choleretic, stimulant, vasodilator, myorelaxant 100 μM
Ursolic acid	Analgesic, antioxidant $IC_{50} = 10$ μM, antiperoxidant $IC_{35} = 200$ μg/mL, protease-inhibitor $IC_{85} = 18$ μg/mL, topoisomerase-II-inhibitor, antiarrhythmic, anticancer, antialzheimer

Adapted from Duke (1992).

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It has also been reported that hyperglycemia is a cause for diabetic complications due to dicarbonyls involved in advanced glycation end product (AGE) formation. Oxidation has been linked to glycation and Mate extracts show a dose-dependent inhibition of dicarbonyl action (Gugliucci and Menini 2002; Lunceford and Gugliucci 2005).

Mate extracts significantly inhibited enzymatic and nonenzymatic lipid peroxidation in rat liver microsomes as well as an effective scavenger of super oxides (Schinella and others 2000). It has been suggested that free radical-induced oxidation of low-density lipoprotein (LDL) plays a role in atherosclerosis. Mate has been shown to inhibit the propagation of LDL oxidation by inhibiting lipid peroxidation as well as DNA oxidation (Gugliucci and Stahl 1995; Gugliucci 1996; Bracesco and others 2003). It has been shown that this mechanism is possible *in vitro*; however, it is still under speculation as to whether it is possible *in vivo*. Evidence also shows that Mate possesses a much higher antioxidant capacity than green tea 13.1 nmol Trolox equivalent antioxidant capacity (TEAC)/μg equivalents gallic acid compared to 9.1 nmol TEAC/μg equivalents gallic acid, respectively (Newell and others 2007).

Weight management and obesity

Obesity is a growing concern in many countries and current research in many areas is directed at finding a way to curb the epidemic. Mate tea has been shown to have possible effects in the area of weight loss and management and current research has provided some supportive evidence. Obese men and women consuming Mate tea have shown a decrease in respiratory quotient (RQ), indicating an increase in fat oxidation (Martinet and others 1999). A herbal infusion made from Mate, guarana, and damiana showed drastic slowing of gastric emptying as well as a decrease in the perceived time for gastric fullness thus increasing satiety. This was also followed by a dramatic decrease in weight, after 45 d, in overweight patients (Andersen and Fogh 2001). Mate has shown to have potential in weight loss and is now being considered as dietary supplement. Adding ingredients such as Mate, guarana, and damiana into supplements has shown to be effective in reducing body weight (Pittler and Ernst 2004). In a randomized, double blind, placebo-controlled clinical trial, Mate was given in a supplement form that also contained green tea, asparagus, black tea, guarana, and kidney bean extracts. The results of this study showed that those taking the supplement had reduced body fat and change in their indexes of body composition (Opala and others 2006). It has been cited that the effect of Mate on weight loss, while not directly known, could be due to its caffeine concentration, contributing to lipolytic activity, or saponin concentration, interfering with cholesterol metabolism and delaying intestinal absorption of dietary fat (Dickel and others 2007). Mate tea can also affect other aspects of lipid metabolism. It has the ability to inhibit atherosclerosis in rabbits when fed with a high cholesterol diet and an aqueous extract of Mate tea (Mosimann and others 2006). Giving Mate extracts to hypercholesterolemic-diet fed rats resulted in a reduction in serum concentrations of cholesterol and triglycerides (Paganini Stein and others 2005). Mate has also shown to have potential as a digestive aid due to a choleretic effect, increasing the rate of bile flow (Gorzalczany and others 2001). One study has also demonstrated that Mate is capable of vaso relaxation of arterial beds in rats. Thus, suggesting that the tea may be able to lower the risk for heart disease, as red wine is believed to do so (Muccillo Baisch and others 1998).

Genotoxic and mutagenic activities

Little data exist regarding the toxicity of Mate tea and standard *in vitro* assays are controversial. In one study, Mate extracts showed to be genotoxic in bacterial cells through induction of functions that regulate responses to DNA damage and disruptions in DNA replication, and mutagenic in *Salmonella typhimurium*. Ames test results showed mutagenic activity at concentrations of 20 to 50 mg aqueous extract/plate and genotoxic at concentrations of 10 to 20 mg aqueous extract/plate. However, when S9 microsomal fraction, catalase, thiourea, or dipyridyl were added to the assay the genotoxic activity of Mate was counteracted, suggesting that oxygen reactive species are the factors responsible for the genotoxicity (Leitao and Braga 1994; Fonseca and others 2000). The results of these *in vitro* tests have not been confirmed in experimental animals or human studies.

Mate association with carcinogenesis

Cancer prevention. *In vitro* and animal experiments have shown a protective effect of Mate against cancer. Several studies have been conducted on the anticancer properties of Mate tea and comparisons have been made with other teas such as green tea, believed to have high anticancer potential (Yamamoto and others 1997). Tests conducted by Ramirez-Mares and others (2004) on *in vitro* chemopreventive activity included cytotoxicity, TPA-induced ornithine decarboxylase (ODC), quinone reductase (QR) activities using HepG2 cells, and topoisomerase inhibitory activity using *Saccharomyces cereviseae*. These tests are of particular importance because cytotoxicity is highly associated with anticancer activity. ODC is a promoter of tumor growth and tumor cells often contain high concentrations of ODC. QR is another screening method for anticancer activity and topoisomerase is required for mitosis; cancer cells show higher concentrations of topoisomerase II (Topo II) than normal cells due to high rates of cell division. Mate was shown to possess the highest cytotoxicity against human liver cancer cells compared to green tea, IC_{50} value of 12.01 g eq. (+) catechin/mL for Mate compared with 72 g eq. (+) catechin/mL for green tea. Table 5 shows the concentrations of tea needed for various inhibitory activities on HepG2 cells.

Human antitopoisomerase II activity was significant and showed a 65% inhibition compared with 15% for green tea (Ramirez-Mares and others 2004). The catalytic topoisomerase inhibition, however, was only on TopoII and not topoisomerase I (Topo I). An *in vitro* study on oral cell carcinoma showed that concentrations greater than 375 μg of solid extract/mL had complete inhibition of cancer cell growth (Gonzalez de Mejia and others 2005). Mate has shown to be a potent TopoII inhibitor and thus showing significant cancer cell growth inhibition, even at low concentrations.

Proteasome inhibitors are an important aspect of cancer research (Osanai and others 2007). The compound epigallocatechin gallate (EGCG), found in green tea, has already been shown to inhibit proteasomes (Osanai and others 2007). Similarly, compounds have been identified in Mate that show proteasome inhibition (Arbiser and others 2005). The compounds identified were 3,5-dicaffeoylquinic acid (3,5-DCQ), 5-caffeoylquinic acid (5-CQ), and 3,4-dicaffeoylquinic acid (3,4-DCQ), which act by inhibiting the chymotrypsin-like activity of a purified 20S proteasome and 26S proteasome in Jurkat

Table 5—Inhibitory effect of Mate tea, green tea, and Ardisia tea against growth of HepG2 cancer cells.[a]

	μg eq. (+) catechin/mL ± SD		
	Mate	Green tea	Ardisia
IC_{10}	9.3 ± 0.6	50.7 ± 2.5	4.9 ± 1.4
IC_{50}	12.0 ± 0.2	72.0 ± 1.8	46.9 ± 3.3
IC_{90}	17.6 ± 0.8	113.6 ± 5.5	177.2 ± 33.4

[a]Adapted from Ramirez-Mares and others (2004).
IC_{10}, IC_{50}, IC_{90} = concentration needed to inhibit 10%, 50%, and 90% cell growth, respectively.
SD = standard deviation.

T (human, peripheral blood, leukemia) cell extracts. Among all of these compounds, 3,5-DCQ showed the highest inhibitory ability. It is believed to act similarly to EGCG due to its similar structure (Arbiser and others 2005).

Other compounds found in Mate have also been studied for their chemopreventive properties. Rutin and quercetin are two that show distinct cytotoxicity to HepG2 cells (Alía and others 2006). Although these compounds are found with small concentrations in Mate they show the diversity of flavonoids present in Mate that contribute to its anticancer potential.

Epidemiological studies. There has been a growing concern over the fact that there are some epidemiological studies that suggest an association between Mate consumption and increased risk of developing certain cancers, namely, esophageal, oral, lung, bladder, renal, and other cancers of the head and neck (Pintos and others 1994; De Stefani and others 1996, 1998; Goldenberg and others 2003; Bates and others 2007). These incidences have been highly correlated to regions in which heavy Mate consumption persists, portions of Brazil and Uruguay. However, it is also recognized that other habitual factors may play a role, such as smoking and alcohol consumption, which are strongly associated with the culture of these regions. Goldenberg (2002) and Goldenberg and others (2003, 2004) report of epidemiological studies showing increased rates of squamous cell carcinoma with increased Mate consumption even when other confounding factors such as smoking were present. The results of these studies indicate that consuming more than 1 L of Mate a day can increase the risk for head and neck cancer by 3 to 5 times, as well as an apparent association to lung cancer (Vassallo and others 1985; De Stefani and others 1996; Sewram and others 2003). It was also reported that consuming strong and very hot tea can increase the risk for oral cancer. Consuming other hot beverages, coffee and green tea, also increased this risk by 2 to 4 times. Thus, the measured risk of oral cancer may be due to thermal injury (Rolon and others 1995; Castellsague and others 2000). With respect to bladder cancer, again epidemiological studies by the same leading authors (De Stefani and others 1991) conducted in Uruguay showed that a relationship between Mate and bladder cancer was found when associated with smoking and to some degree in nonsmokers as well, though less defined. In the same study, it was also shown that consumers of black tea and coffee had an increased risk of bladder cancer. An epidemiological study conducted in Argentina showed an increase risk of bladder cancer in Mate drinkers and smokers but not in nonsmokers (Bates and others 2007). Whether this increased risk of bladder cancer is due solely to Mate alone, smoking alone, a combination of both, or solely another cause is unclear.

It should also be noted that the case studies of Mate consumption and increased cancer incidence also include individuals that consume black tobacco and alcohol, namely, wine. De Stefani and others (1988) stated that there is a correlation to the increased risk of oral cancers in those individuals who consume wine, Mate, and smoke. This increase is also noted to be greater in those who smoke black tobacco over blond tobacco. Again, there is no direct implication that any one factor contributes more to this increase in oral cancers. Due to these other confounding factors, Mate may not be a carcinogen on its own but, due to the high temperature at time of consumption, may in fact be a means of increasing absorption for the carcinogens found in cigarette smoke and other environmental contaminants that are carcinogens or cancer promoters (Goldenberg and others 2004).

On the other hand, there may be compounds present in Mate that could contribute to cancer. Fagundes and others (2006) have shown a correlation between the amount of Mate consumed and the amount of polycyclic aromatic hydrocarbons (PAHs) in the body. It is known that PAHs, particularly benzo[a]pyrene, have carcinogenic properties and that tobacco smoke and grilled meat contain high concentrations of PAHs; at least 15 PAH compounds have been found in Mate varieties. These compounds were isolated and identified by the utilization of stir bar sorptive extraction (SBSE) and high-performance liquid chromatography–fluorescence detection (HPLC–FLD) (Zuin and others 2005). Total PAHs found in various Brazilian Mate samples ranged from 600 to 2300 ng/L, with naphthalene, acenaphthene, and phenanthrene having the highest concentrations. Table 6 shows the PAH compounds identified in Mate and their average concentration in 11 Mate samples.

It is known that exposure to PAHs through tobacco smoke and other sources may increase the risk of esophageal squamous cell carcinoma (ESCC). Fagundes and others (2006) evaluated 200 healthy adult Mate tea consumers, half male and half female with half being smokers and half being nonsmokers, to determine the concentrations of 1-hydroxypyrene glucuronide (1-OHPG), a PAH glucuronide detoxification metabolite excreted in the urine. Their presence provides evidence that an individual has been exposed to PAHs. 1-OHPG can be measured in the urine using immunoaffinity chromatography, synchronous fluorescence spectroscopy, and a urine cotinine dipstick test; the tests were conducted by the Natl. Cancer Inst. at Johns Hopkins Univ. This study found that there was a direct correlation between the amount of Mate consumed and the concentrations of PAHs in the urine, the higher the consumption the higher the concentrations. Table 7 shows the increasing concentrations of 1-OHPG in the urine with increasing Mate consumption.

However, other than an increase in Mate consumption alone, higher concentrations of 1-OHPG can also be correlated with a combination of smoking and Mate drinking. When Mate consumption is combined with smoking, 1-OHPG concentrations are significantly higher but Mate alone produces about the same concentrations of 1-OHPG on average as smoking alone (Fagundes and others 2006). When examining a population in Campinas, SP, Brazil and the coffee and Mate they consumed, PAHs were found in all products and ranged from 10.12 μg/kg for coffee to 0.70 μg/kg for Mate (Rojo de Camargo and others 2002). Considering the per capita average daily consumption estimates in Brazil of 69.79 g of Mate tea, one can assume that Mate tea contributes with approximately 0.05 μg of total PAHs to the dietary intake of these contaminants by the studied population ($n = 600$) (Rojo de Camargo and others 2002).

Table 6—Average concentration of polycyclic aromatic hydrocarbons found in Brazilian Mate tea samples.[a]

Compound	ng/L	Compound	ng/L
Acenaphthene	426.3	Benzo(b)fluoranthene	11.4
Phenanthrene	347.5	Chrysene	10.5
Naphthalene	96.5	Benzo(a)anthracene	9.7
Fluoranthene	61.4	Indeno(1,2,3)pyrene	9.5
Pyrene	59.1	Benzo(g,h,i)perylened	7.7
Anthracene	50.9	Dibenz(a,h)anthracene	5.0
Fluorene	29.7	Benzo(k)fluoranthene	3.6
Benzo(a)pyrene	12.2		

[a]Adapted from Zuin and others (2005).

Table 7—Concentration of 1-hydroxypyrene glucuronide (1-OHPG) in urine of humans.[a]

Mate consumption (mL/day)	1-OHPG (pmol/mL)
<100	1.01
>100	1.97
>500	3.24
>1000	4.06

[a]Adapted from Fagundes and others (2006).

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Although there has been no proven biological correlation to the drinking of Mate and developing cancer (Pereira Jotz and others 2006), the contamination with PAHs does present a plausible explanation for increased rates of Mate drinking and cancer. It is highly probable that PAHs are obtained in the processing, as Mate is commonly dried over a smoky wood fire. The smoke from the wood may thus be producing the PAHs found in Mate. There also appears to be an apparent lack of new information on the subject. Though a number of papers are published on the topic, no new evidence has been presented. This is an area that warrants further investigation.

Technological Considerations

Flavor and aroma

Consumer preference and perception are key attributes to any food product and the same can be said for Mate tea. The driving force behind sales and brand selection and consumer preference for Mate brands is largely driven by smell and taste attributes. Generally sensory panels do the analysis of these characteristics; however, it is expensive and subject to susceptibility of the panelists. Therefore, an automated method for aroma determination is needed. Grigioni and others (2004) showed that the use of an E-nose can discriminate among the aroma characteristics of Mate and correlates to that of trained panelists.

It has been shown that there is a direct correlation between the consumer preference for taste and aroma to the appearance of the product (Cruz and others 2003; Schneider and others 2006). When sensory panels are used, key terms must be generated to define the flavor, aroma, and appearance of Mate products. Descriptors of these characteristics that have shown to differentiate products are shown in Table 8 (Santa Cruz 2002; Cruz and others 2003). Consumer panelists have also been used to test the bitterness (Calvino and others 2004).

The aroma compounds found in Mate have also been characterized using gas chromatography/mass spectrometry; while not correlated to sensory analysis it does show the chemical make-up of the volatile constituents of Mate. It was shown that Mate contains more than 250 components, many of which are the same as to green tea. However, a number of distinct components were identified, namely, 2-butoxy-ethanol (in high concentrations), and 3,3,5-trimethylcyclohexanone-related compounds. Among the 196 volatile chemical compounds found in Yerba Mate tea, only 144 are present in green tea (Kawakami and Kobayashi 1991).

Mate tea infusions can be made from green Mate, the dried ground leaves, or roasted green Mate, where the dried leaves are further roasted to enhance flavor. This roasting process has been shown to have a dramatic effect on the flavor and aroma of the tea. Numerous studies have been conducted to examine the volatile compounds found in Mate. Roasted Mate showed higher concentrations of furans, pyrazines, and pyrroles compared to green Mate, likely due to Maillard reactions (Kawakami and Kobayashi 1991). Bastos and others (2006b) examined the essential oil extracts from green and roasted Mate and found that roasted Mate contained significantly less of the compounds responsible for the green-floral aroma, that is, limonene, which are characteristics of green Mate. They also found an increase in compounds such as methyl furfural and furfural, which may be responsible for the smoky characteristics of roasted Mate. Table 9 shows the volatile compounds found in green and roasted Mate tea in comparison to black tea, identified with aroma analysis using solvent-assisted flavor evaporation–solvent extraction (SAFE–SE). (Kawakami and Kobayashi 1991).

Lozano and others (2007) used 3 different methods to determine the volatile aroma compounds present in Mate. SAFE–SE analysis identified the highest number of compounds followed by adsorptive column extraction with aroma extract dilution analysis (ACE–AEDA) and dynamic headspace dilution analysis (DHDA), which found a similar number of compounds. However, each method identified compounds that were not identified with another method. Therefore, it is recommended that multiple methods be used for the analysis of volatile aroma compounds. Table 10 presents the main aroma compounds and their characteristic odor identified with 3 different methods for 1 Mate tea.

One of the defining characteristics of Mate teas is the perception of bitterness. This characteristic can be attributed to caffeine (Ley and others 2006; Keast and Roper 2007) as well as tannins (Drinkine and others 2007) and saponins (Ma and others 1989). It should be noted that the presence of stems, often found in most varieties, could significantly reduce the concentration of bitterness compared with those without stems (Calvino 2005).

Table 8 — Yerba Mate sensory descriptors.[a]

Appearance of dry Mate	Appearance of Mate infusion	Flavor and aroma
Stick and leaf size	Sediment	Initial impact
Stick and leaf size uniformity	Turbidity	Acid
Quantity of sticks	Brown color	Humid
Quantity of dust		Smoke
		Paper
		Chemical
		Green
		Toasted
		Residual
		Bitterness

[a]Adapted from Cruz and others (2003).

Compound extraction

While Mate is primarily consumed in a beverage form, made by steeping the leaves of the plant in hot water, its high concentrations of beneficial compounds make it an interesting subject for extraction and purification of these compounds for use in the nutraceutical industry. The use of sonication has been shown to effectively eliminate high concentrations of compounds from Mate, that is, caffeine and theobromine. However, this method is affected by solvent polarity as well as extraction time and solvent to sample mass ratio (Jacques and others 2007). The sonication method also requires the use of organic solvents, methanol, and hexane, which can be troublesome when the extracts are to be used for human consumption. Because of this, supercritical CO_2 extraction appears to be more promising for this extraction purpose. By utilizing supercritical CO_2 extraction, the concentrations of methylxanthines extracted are much higher compared to other extraction methods. The use of supercritical CO_2 was investigated and was found to be an effective extraction method for caffeine, with yields of 98% total caffeine. This method also showed that it is possible to extract theobromine. It was also shown that supercritical CO_2 has a higher affinity for caffeine than theobromine. When ethanol is utilized in the extraction as well, extraction efficiency was improved by lowering solvent and energy requirements (Saldana and others 1999; Saldana and others 2002).

The use of supercritical CO_2 has now been employed for the analysis of Mate samples as a determination of quality differences. Samples of Mate were tested using CO_2 extraction to examine changes in the concentrations of caffeine, theobromine, phytol, vitamin E, squalene, and stigmasterol due to differences in light exposure, drying method, and age of leaves (Esmelindro and others 2004). The

data showed that when products were protected from light there was a dramatic increase in concentrations of caffeine, theobromine, phytol, and on the steroid stigmasterol, especially caffeine and theobromine, roughly 3 times higher. Light exposure appears to have no effect on vitamin E concentration. Age of leaves played a role in the amount of all compounds; younger leaves showed the highest concentrations of all compounds. When alternative methods to air drying were used, microwave drying allows for the greatest retention of compounds compared to vacuum drying. These findings are significant because they show that light conditions during growing, age of leaves, and drying method may play a role in the composition of Mate and this would be important in the selection of products for extraction in producing a high-quality extract (Esmelindro and others 2004).

Final Considerations

When comparing Mate to other teas such as green tea and black tea, several differences can be observed. Most notably the flavor and aroma, distinctly bitter Mate is often characterized as an acquired taste. The roasted/smoky aromas are also often a much-desired characteristic and ones that distinguish it from other teas. It is not only the outward properties that distinguish it from other teas but also its diverse concentration of biological compounds that are not readily found in other teas. Most notably of these compounds are the xanthines, theobromine, and theophylline that are attributed to its ability to increase energy levels. The saponin concentration is also noteworthy in that they are not found with high concentrations in other teas; the saponins contribute to the flavor and may also be attributed to anti-inflammatory and hypocholesterolemic properties characteristic to Mate as a medicinal herb. It should also be mentioned that, though Mate is high in many compounds not found in other teas, it does not contain catechins like green tea and is not as high in flavonoids as black tea.

Most notably of Mate's biological activities is its high antioxidant capacity which has been shown to be higher than green tea, which is touted as having a very high antioxidant capacity. This high antioxidant capacity is attributed and is directly proportional to its high polyphenol concentration, namely, the caffeoyl derivatives. Due to Mate's high biological activity and its large concentration of known active compounds it makes an ideal material for extraction of these compounds for use in other foods and supplements. There are currently several products in the market that contain some derivatives of Mate. Most of which are targeted at weight loss, as Mate has shown a correlation with weight loss and weight management. Future research will likely show more precise mechanisms for Mate's actions in these areas.

Contrary to the reported carcinogenic properties of Mate, there are scientifically backed reports of anticancer effects. Mate tea has been shown to have a high cytotoxicity for cancer cells, which is even higher than that of green tea. Mate has also shown to be highly effective in inhibiting topoisomerase II, which is responsible for cell division and by inhibiting cancer cell proliferation. It has been shown that oral cancer cells can be completely inhibited by treating them with 375 μg of Mate extract/mL. It should also be noted that, though Mate does not contain catechins, that is, EGCG, it does have compounds that act similarly, such as 3,5-dicaffeoylquinic acid. This compound has shown to be a potent proteasome inhibitor comparable to EGCG, which has known proteasome inhibition activity and is being investigated for cancer treatment.

Table 9 — Volatile compounds in green Mate and roasted Mate compared to *Camellia sinensis* tea (black tea).[a]

Compound	Green Mate	Roast Mate	Black tea	Compound	Green Mate	Roast Mate	Black tea
(E)-2,(E)-4-heptadienal	●	●	●	dihydroactinidiolide	●	●	●
(E)-2-decenal	●	●	●	eugenol		●	
(E)-2-hexenal	●	●	●	furfural	●	●	●
(E)-2-pentenal	●	●	●	furfuryl alcohol	●	●	●
(E)-2-pentenol	●	●	●	geranial	●	●	●
(E)-2-undecenal	●	●	●	geraniol	●	●	●
(E)-3,(2)-5-octadien-2-one	●	●	●	geranylacetone	●	●	●
(E)-3,(E)-5-octadien-2-one	●	●	●	guaiacol	●	●	●
1,3,5-trimethyl-2-(1,3-butadienyl)benzene	●	●		heptanoic acid	●	●	●
2,10,10-trimethyl-6-methylidene-l-		●		heptanol	●	●	●
2,3-dihydro-2-methylbenzofuran	●	●		hexanal	●	●	●
2,6,6- trimethyl-2- hydroxycyclohexanone	●	●	●	hexanoic acid	●	●	●
2,6,6-trimethylcyclohex-2-enel,-4 –dione	●	●	●	l-penten-3 — 01	●	●	●
2-acetylfuran	●	●	●	l-phenylpropanone	●	●	●
2-butoxyethanol	●	●		limonene	●	●	●
2-decanone	●	●	●	linalool	●	●	●
2-ethylfuran	●	●	●	linalool oxide I (cis, furanoid)	●	●	●
2-methyl-2-pentenal	●	●		linalool oxide II (trans, furanoid)	●	●	●
2-methyl-3-buten-2–01	●	●	●	linalool oxide III (cis, pyranoid)		●	●
2-methylbutanoic acid	●	●	●	linalool oxide IV (trans, pyranoid)	●	●	●
5,6-epoxy-ionone	●	●	●	methyl salicylate	●	●	●
5-methylfurfural	●	●	●	nerol	●	●	●
6,10,14-trimethylpentadecanone	●	●	●	nonanoic	●	●	●
6-methyl-(E)-3,5-heptadien-2-one	●	●	●	o-cresol	●	●	●
6-methyl-S-hepten-2-one	●	●	●	octanoic acid	●	●	●
acetic acid	●	●	●	octanol	●	●	●
a-ionone	●	●	●	pentanal	●	●	●
a-terpineol	●	●	●	pentanol	●	●	●
Benzaldehyde	●	●	●	phenol	●	●	●
benzyl alcohol	●	●	●	propionic acid	●	●	●
butyric acid	●	●	●	valeric acid	●	●	●
decanoic acid	●	●	●	β-ionone	●	●	●

[a]Kawakami and Kobayashi (1991).
Compounds are 0.5% or more of total concentration.

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Table 10—Main volatile aroma compounds in Mate found by 3 different analytical methods.[a]

Compound	SAFE	ACE-AEDA	DHDA	Odor
(E)-2-Decenal	•		•	Green, pungent
(E)-2-Nonenal	•		•	Hay
(E)-2-Octenal	•	•	•	Raw peanut
(E,E)-2,4-Hexadienal	•			Fatty, metalic
(E,Z)-2,6-Nonadienal	•	•	•	Cucumber
(Z)-1,5-Octadien-3-one			•	Metalic
(Z)-2-Nonenal			•	Melon, hay
(Z)-3-Hexenal	•			Green, cut-leaf
(Z)-4-Heptenal	•		•	Rancid
1,8-Cineole	•	•		Minty, eucalyptus
1-Hexen-3-one	•		•	Plastic
1-Oten-3-ol	•	•	•	Mushroom
1-Penten-3-one	•	•	•	Plastic, rancid
2,3-Butanedione	•		•	Buttery, creamy
2,3-methylbutanal	•			Chocolate
2,3-Pentanedione	•			Buttery, creamy
2-Acetyl-1-pyrroline	•	•	•	Roasty, popcorn
2-Acetyl-2-thiazoline		•		Roasty, popcorn
2-Acetylthiazole	•	•		Roasty, popcorn
Butanoic acid	•	•	•	Sweaty, cheesy
Citronellol	•		•	Fruity
Eugenol	•	•	•	Cloves, spicy
Furaneol	•			Burnt sugar
Geranial	•	•	•	Fruity, floral
Geraniol	•	•	•	Floral
Guaiacol	•	•	•	Smoky, medicine
Hexanal			•	Green, cut-grass
Hexanoic acid	•	•		Sweaty, body odor
Linalool	•	•	•	Floral
Maltol		•		Burnt sugar
Methional	•	•	•	Cooked potato
β-Damascenone	•			Cooked apple
Nonalactone	•	•	•	Coconut, sweet
o-Cresol		•		Phenolic, medicine
Octalactone	•			Fruity, floral
Octanal			•	Orange oil
p-Cresol	•	•		Phenolic, animal, dung
Pentanoic acid	•			Sweaty, cheesy
p-Vinyl guaiacol	•	•		Cloves, spicy
Skatole	•	•		Urine, mothballs
Wine lactone	•			Plastic
β-Damascenone		•	•	Cooked apple
β-Ionone	•	•	•	Floral

[a]Adapted from Lozano and others (2007).

Conclusions

Yerba Mate has been consumed for centuries but it has only been scientifically studied in the last 2 decades. The growing worldwide interest in Mate has made it paramount that research on this herbal tea continues, as it has shown extraordinary possibilities not only as a consumer beverage but also in the nutraceutical industry. In regard to carcinogenesis, the most recent information suggests that the association between Mate consumption and the occurrence of cancer may not be due to raw Mate itself but to contaminants that may be present in processed Mate. The high temperature at which Mate tea is consumed may also play a role. Therefore, postharvest technologies need to be improved—especially the drying process needs to be optimized to completely eliminate contaminants. Additionally, good quality control, including throughout analytical testing, becomes imperative to insure its safety.

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